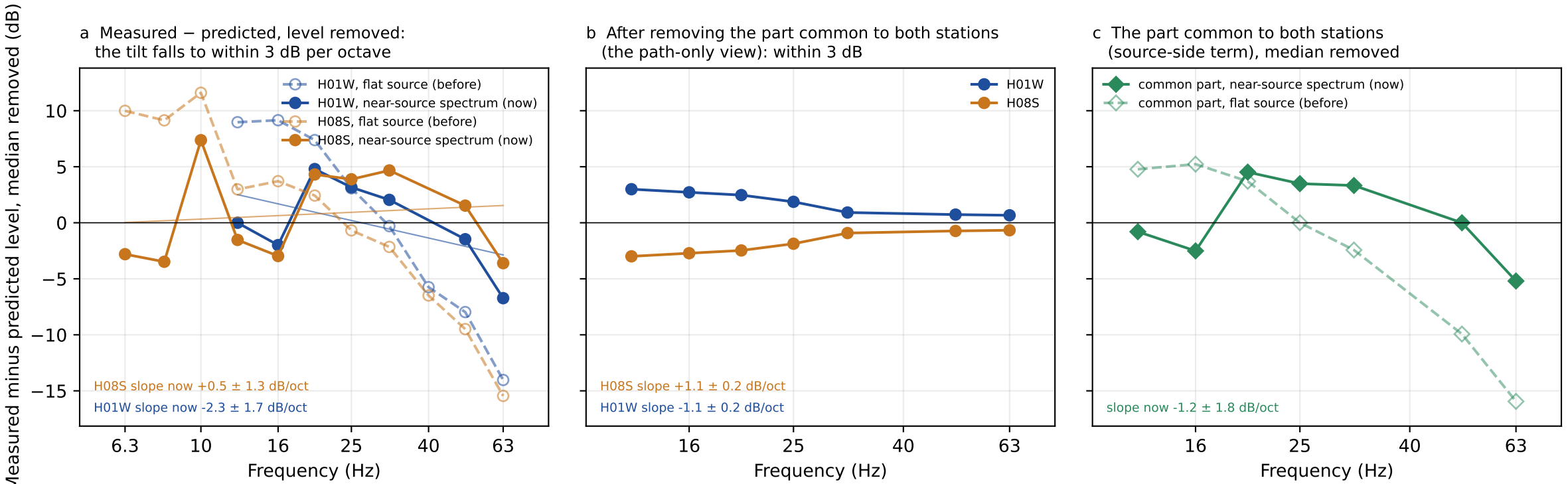
Airgun shots of October 2001 (site air9): the frequency tilt of the misfit, before and after the source spectrum from the ship's near-source recording



What the chart shows: how far the predicted sound level is from the measured one at each frequency, at two listening stations. (a) With a flat source the misfit tilted strongly with frequency (dashed); with the measured near-source spectrum the tilt is within 3 dB per octave (solid). (b) The part of the misfit that differs between the two stations is within 2 dB. (c) The part shared by both stations, which is the source-side term, is now flat to about 1 dB per octave. The vertical offset is not shown (it mixes a peak level with a spectrum level). Caveats C1 and C2 apply.

Technical: r_b = RL_obs − RL_pred (dB; audit sign reversed: here r > 0 means the model under-predicts the signal), nominal geometry (z_s 10 m, Ball 30 m ahead, 15 m deep, OFR 95-228 layout, V^(1/3) weights, SL_B 235). 'Before' = −(TL_obs − TL_model) of audit M1 (flat source). Common mode c_b = mean of r_b over the two stations on the 7 shared bands (12.5–63 Hz, 40 Hz excluded); r~_b = r_b − c_b. OLS slopes on log2 f with standard errors; grades per criteria.md 940f1a3. Sources DIGITISED: Blackman 2004 Fig. 23, EW0112 Fig. 2. Pre-registration e949144; verdict NOT EXPLAINED (level test). No MH370 posterior enters; C1 (slope misfit) and C2 (η not calibrated) stay on every MH370 number.